

Interpretable and Tradable Factors from Mutual Funds via Deep Autoencoding with Alternative Data

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Abstract

Factor models are central to asset pricing and portfolio construction, yet dominant models such as Fama-French are designed for equities and explain little of the return variation in bonds, REITs, and emerging market assets. Deep latent factor models improve in-sample fit but typically produce abstract, non-tradable factors that resist economic interpretation. We propose the Autoencoding Factor Model (AFM), a neural autoencoder whose bottleneck layer is a set of explicit portfolio returns: each latent factor is the return of a tradable portfolio whose weights are generated by a neural network conditioned on mutual fund characteristics—sector holdings and asset class allocations from regulatory filings—and a learned market-state embedding from past returns. Trained end-to-end on approximately 1,000 US mutual funds spanning 10 asset classes, the AFM with five factors achieves 85.0% average out-of-sample R^2 , a 16.6 percentage point improvement over the Fama-French four-factor model. The largest gains arise on non-equity assets where traditional factors have near-zero explanatory power: US aggregate bonds (84% vs. 11%), emerging market debt (44% vs. 11%), and high yield (53% vs. 35%).

1 Introduction

Factor models are the workhorse of asset pricing, risk management, and portfolio construction. Practitioners and academics rely on a small number of common factors—market, size, value, momentum—to explain the cross-section of returns, allocate risk budgets, and evaluate investment performance. Yet the dominant factor models are built from equity sorts and evaluated almost exclusively on stock returns. Bonds, REITs, and emerging market assets collectively represent trillions of dollars in assets under management, and for these asset classes the standard Fama-French factors explain as little as 11% of return variation out of sample.

Deep learning has transformed factor modeling in recent years. Conditional autoencoders learn nonlinear mappings from asset characteristics to time-varying factor loadings [15], while GAN-based stochastic discount factor estimation [6], risk-premium PCA [22], and instrumented PCA [19] push predictive performance further on US equities. VAE-based extensions introduce probabilistic factor structure [9], and attention mechanisms capture cross-asset interactions [11]. These models achieve impressive out-of-sample R^2 and Sharpe ratios on individual stock returns.

Two critical gaps remain, however. First, most deep factor models produce latent factors that are abstract statistical objects—they cannot be directly traded or economically interpreted. While Gu et al. [15] originally proposed a no-arbitrage weight structure, subsequent extensions relax this constraint in favor of richer latent representations, severing the link between learned factors and implementable trading strategies. Second, virtually all deep factor models are trained and evaluated exclusively on US equities. Whether these architectures generalize to bonds, REITs, and emerging market assets—where the underlying return drivers differ fundamentally from equity risk premia—is unknown. Meanwhile, fund-level alternative data such as portfolio holdings and sector allocations from regulatory filings remains unexploited as conditioning information for factor construction.

We propose the Autoencoding Factor Model (AFM), which addresses both gaps simultaneously. The key architectural innovation is a *portfolio bottleneck*: rather than learning free latent codes, the AFM constrains each factor to be the realized return of an explicit portfolio whose weights are produced by

a neural network and mapped to valid allocations via softmax (long-only) or a softmax-minus-softmax construction (market-neutral). Factor loadings and intercepts are generated by separate networks conditioned on fund-level characteristics—economic sector holdings and asset class allocations derived from SEC N-PORT filings—together with a learned market-state embedding that compresses a 252-day return window. Because all portfolio weights and loadings depend only on information available at time $t - 1$, every factor is tradable without look-ahead bias. The entire system is trained end-to-end via return reconstruction loss, inheriting the scalability of modern deep learning while producing factors that connect directly to the classical tradition of mimicking portfolios [16] and parametric portfolio policies [4].

Contributions.

- **Portfolio bottleneck architecture.** We propose a neural autoencoder where each latent factor is constrained to be the return of a tradable portfolio with explicit, no-look-ahead weights, bridging deep latent factor models [15] and classical portfolio-based factor construction [16]. This is, to our knowledge, the first autoencoder factor model that enforces tradability as a hard architectural constraint rather than a soft regularizer.
- **Conditional factor structure via fund-level alternative data.** Portfolio weights and factor loadings are conditioned on mutual fund characteristics (economic sector holdings, asset class allocations) derived from regulatory filings, enabling interpretable, time-varying factor exposures that reflect the economic composition of each fund.
- **Cross-asset factor learning.** Trained on approximately 1,000 US mutual funds across 10 asset classes—spanning large, mid, and small cap equities, international and emerging market equities, US aggregate and high-yield bonds, emerging market debt, municipal bonds, and REITs—the AFM achieves 85.0% average out-of-sample R^2 compared to 68.4% for the Fama-French four-factor model [5, 12], with the largest gains on non-equity assets where traditional equity factors have near-zero explanatory power.
- **Comprehensive ablation study.** We systematically evaluate architecture choices (MLP vs. residual MLP), training universes (single-asset-class vs. multi-asset), and their interactions. We find that simpler architectures and broader training universes yield superior generalization, with multi-asset training improving average R^2 by 12 percentage points over large-cap-only training.

Paper Organization. The remainder of this paper is organized as follows. Section 2 reviews related work on deep factor models, autoencoder extensions, and fund-level machine learning. Section 3 formalizes the AFM architecture and training procedure. Section 4 presents our main results and ablation studies. Section 5 concludes.

2 Related Work

2.1 Latent Factor Models and Deep Asset Pricing

Classical factor models originate from Ross [23]’s Arbitrage Pricing Theory, which establishes that expected returns are linearly related to exposures to a small number of common factors. The empirical implementations by Fama and French [12] and Carhart [5] construct tradable factors via characteristic sorts, but are designed for equities and have limited explanatory power for bonds, REITs, and emerging market assets. Statistical approaches such as PCA [3] and RP-PCA [22] recover latent factors that maximize explained variation or penalize pricing errors, while IPCA [19] instruments loadings with observable characteristics, enabling time variation within a linear framework. More recently, Gu et al. [15] introduced a conditional autoencoder that maps characteristics to factor loadings via neural networks, and subsequent work has extended this paradigm: Chen et al. [6] adopt adversarial GAN-based SDF estimation with LSTM-extracted macro states, Epstein et al. [11] apply attention mechanisms to construct conditional arbitrage factors, and Kelly et al. [20] develop a Portfolio Tangent Kernel decomposition for scaling deep factor models. Unlike these works, which operate on individual stock returns and produce abstract latent factors that cannot be directly traded, our model operates on mutual fund returns and constrains each factor to be the return of an explicit, tradable portfolio via the portfolio bottleneck.

2.2 Autoencoder and VAE Extensions

A growing body of work extends the conditional autoencoder architecture along several dimensions. Duan et al. [9] introduce FactorVAE, a variational framework that models latent factors as distributions rather than point estimates, enabling uncertainty quantification in noisy financial data. Wang and Guo [26] extend this with recurrent encoders to capture temporal factor dynamics, while Wang and Singh [25] replace standard MLP layers with Kolmogorov-Arnold networks for improved interpretability of learned loadings. Gopal [13] propose NeuralFactors, which jointly learns factor exposures and factor returns within a VAE framework, and Engel et al. [10] scale conditional autoencoders to high-dimensional factor spaces with uncertainty-driven pruning. Jang et al. [17] take a different approach entirely, using analyst consensus forecasts as a structural bottleneck to inject domain knowledge into the latent space. Unlike these extensions, which focus on probabilistic modeling, alternative network architectures, or information-theoretic bottlenecks—all applied to individual stock returns—our bottleneck is a *portfolio return*: a fundamentally different inductive bias that enforces tradability by construction and operates at the fund level with fund-specific conditioning variables.

2.3 Portfolio Construction and End-to-End Learning

The idea that factors should correspond to tradable portfolios dates to Huberman et al. [16], who showed that any factor can be replicated by a portfolio that is maximally correlated with it. Brandt et al. [4] formalize parametric portfolio policies where weights are linear functions of asset characteristics, enabling direct optimization of portfolio objectives. A parallel thread in decision-focused learning differentiates through optimization layers: Amos and Kolter [2] embed quadratic programs as differentiable network layers, Donti et al. [8] propose task-based end-to-end learning that minimizes downstream decision loss, and Zhang et al. [27] optimize Sharpe ratios directly via differentiable portfolio construction. Unlike these approaches, which learn portfolio allocations to maximize trading performance, our model learns factor-mimicking portfolios that maximize cross-sectional explanatory power—bridging factor modeling and portfolio construction in a single architecture.

2.4 Machine Learning for Mutual Funds

Sharpe [24] introduced returns-based style analysis, decomposing fund returns into exposures to asset class indices via constrained regression. Recent work applies modern ML to fund analysis: Kaniel et al. [18] train neural networks on stock, fund, and macroeconomic characteristics to identify skilled fund managers, while DeMiguel et al. [7] demonstrate that ML methods exploiting a large set of fund characteristics can generate significant out-of-sample alpha. Unlike these approaches, which use fund characteristics to predict fund performance or select funds, we use fund-level characteristics derived from regulatory filings—sector holdings and asset class allocations—as *conditioning variables for factor construction*. This is a fundamentally different task: rather than predicting which funds will outperform, we learn a latent factor structure that explains *why* fund returns co-move across asset classes.

Positioning. To our knowledge, no prior work combines (i) a portfolio bottleneck that enforces factor tradability by construction, (ii) conditioning on fund-level alternative data from regulatory filings, and (iii) multi-asset training across equities, bonds, REITs, and emerging markets within a unified autoencoder framework. Our work bridges deep latent factor models (§2.1–2) with portfolio-based factor construction (§2.3), applied to the mutual fund universe (§2.4).

3 Method

We present the Autoencoding Factor Model (AFM), a neural architecture that learns tradable factor portfolios from mutual fund returns and fund-level characteristics. The key innovation is a *portfolio bottleneck*: each latent factor is defined as the return of an explicit portfolio whose weights are produced by a neural network using only past information, guaranteeing tradability by construction. We formalize the problem setup (§3.1), describe the architecture (§3.2–§3.3), present the training objective (§3.4), and discuss the role of the portfolio bottleneck (§3.5).

Table 1: Notation summary.

Symbol	Meaning
N, T, F	Number of funds, time periods, latent factors
D, S, L	Characteristic dim., state dim., lookback length
$r_{i,t}, \mathbf{r}_t$	Excess return of fund i ; N -vector of returns
$\mathbf{x}_{i,t}, \mathbf{X}_t$	Characteristic vector of fund i ; $N \times D$ matrix
$\mathbf{R}_{t-1}^{(L-1)}$	$N \times (L-1)$ lagged return panel
$\mathcal{F}_{t-1}$	Information set at $t-1$
$\mathbf{s}_{t-1}$	Learned market state vector ($\in \mathbb{R}^S$)
$\mathbf{W}_{t-1}, \mathbf{w}_{t-1}^{(k)}$	$N \times F$ weight matrix; weights for factor k
$\mathbf{f}_t$	F -vector of realized factor returns
$\mathbf{B}_{t-1}, \beta_{i,t-1}$	$N \times F$ loading matrix; loadings for fund i
$\boldsymbol{\alpha}_{t-1}$	N -vector of asset intercepts
$g_\psi, h_\phi, b_\theta, a_\eta$	ReturnEncoder, ScoreNet, BetaNet, AlphaNet
k_0	Number of long-only factors

3.1 Problem Setup

We observe a panel of excess returns for N mutual funds over T trading days. Let $r_{i,t} \in \mathbb{R}$ denote the excess return of fund i at time t , and stack these as $\mathbf{r}_t = (r_{1,t}, \dots, r_{N,t})^\top \in \mathbb{R}^N$. Each fund i has an observable characteristic vector $\mathbf{x}_{i,t} \in \mathbb{R}^D$ derived from regulatory filings (economic sector holdings and asset class allocations), with the cross-section stacked as $\mathbf{X}_t \in \mathbb{R}^{N \times D}$. Fix a lookback window of length L ; the lagged return panel available at time $t-1$ is

$$\mathbf{R}_{t-1}^{(L-1)} := (\mathbf{r}_{t-L+1}, \dots, \mathbf{r}_{t-1}) \in \mathbb{R}^{N \times (L-1)}. \quad (1)$$

The information set at time $t-1$ is $\mathcal{F}_{t-1} := \sigma(\mathbf{X}_{t-1}, \mathbf{R}_{t-1}^{(L-1)})$. All quantities used to form portfolios at time t are functions of $\mathcal{F}_{t-1}$, ensuring **no look-ahead**.

Goal. Learn F latent factors that admit a conditional linear factor model

$$\mathbf{r}_t \approx \boldsymbol{\alpha}_{t-1} + \mathbf{B}_{t-1} \mathbf{f}_t, \quad (2)$$

where $\boldsymbol{\alpha}_{t-1} \in \mathbb{R}^N$ is a vector of asset intercepts, $\mathbf{B}_{t-1} \in \mathbb{R}^{N \times F}$ is a matrix of conditional factor loadings, and $\mathbf{f}_t \in \mathbb{R}^F$ is a vector of realized factor returns—all with $\boldsymbol{\alpha}_{t-1}$ and $\mathbf{B}_{t-1}$ measurable with respect to $\mathcal{F}_{t-1}$. Crucially, we require that factors are *tradable portfolios*:

$$\mathbf{f}_t := \mathbf{W}_{t-1}^\top \mathbf{r}_t, \quad \mathbf{W}_{t-1} \in \mathbb{R}^{N \times F}, \quad (3)$$

where $\mathbf{W}_{t-1}$ is $\mathcal{F}_{t-1}$ -measurable. Table 1 summarizes the notation used throughout.

3.2 Architecture Overview

The AFM has the structure of an autoencoder whose bottleneck is a vector of portfolio returns (Figure 1). The *encoder* maps the information set $\mathcal{F}_{t-1}$ to portfolio weights $\mathbf{W}_{t-1}$ and factor exposures $(\boldsymbol{\alpha}_{t-1}, \mathbf{B}_{t-1})$. The *bottleneck* computes factor returns $\mathbf{f}_t = \mathbf{W}_{t-1}^\top \mathbf{r}_t$ as realized portfolio returns. The *decoder* reconstructs fund returns via $\hat{\mathbf{r}}_t = \boldsymbol{\alpha}_{t-1} + \mathbf{B}_{t-1} \mathbf{f}_t$.

Four sub-networks parameterize this mapping:

1. **ReturnEncoder** (g_ψ): compresses lagged returns into a market state vector;
2. **ScoreNet** (h_ϕ): produces portfolio scores, mapped to valid weights via softmax;
3. **BetaNet** (b_θ): outputs conditional factor loadings per fund;
4. **AlphaNet** (a_η): outputs fund-specific intercepts.

All four networks receive only $\mathcal{F}_{t-1}$ -measurable inputs, so every output—weights, loadings, and intercepts—satisfies the no-look-ahead requirement by construction. We describe each sub-network next.

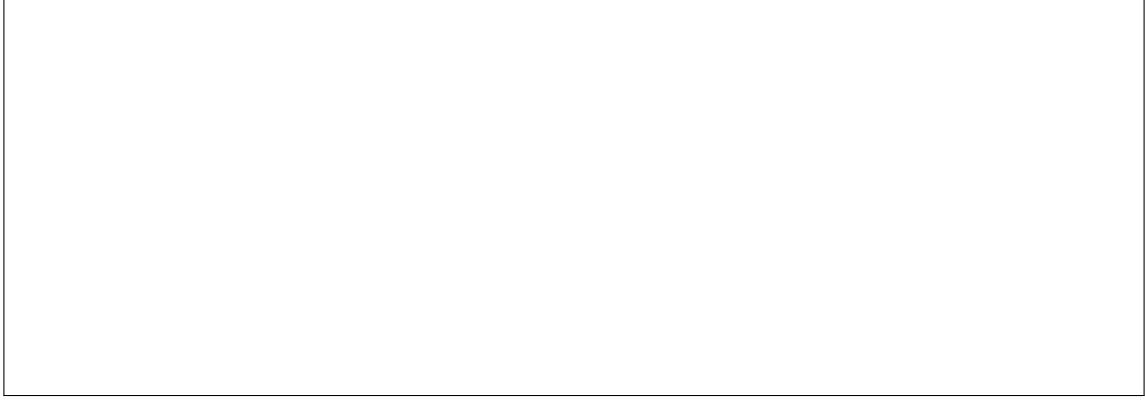


Figure 1: **AFM architecture overview.** The encoder maps fund characteristics and a learned market-state embedding to portfolio weights (ScoreNet), factor loadings (BetaNet), and intercepts (AlphaNet). The bottleneck computes factor returns as explicit portfolio returns $\mathbf{f}_t = \mathbf{W}_{t-1}^\top \mathbf{r}_t$. The decoder reconstructs fund returns via $\hat{\mathbf{r}}_t = \boldsymbol{\alpha}_{t-1} + \mathbf{B}_{t-1} \mathbf{f}_t$. All weights and loadings depend only on $\mathcal{F}_{t-1}$, ensuring no look-ahead.

3.3 Sub-Network Details

ReturnEncoder. Financial cross-sections are shaped by time-varying market conditions—volatility regimes, interest rate environments, and risk appetite. The ReturnEncoder compresses the lagged return panel into a low-dimensional state vector:

$$\mathbf{s}_{t-1} = g_\psi \left(\text{vec} \left(\mathbf{R}_{t-1}^{(L-1)} \right) \right) \in \mathbb{R}^S, \quad (4)$$

where g_ψ is an MLP with layer normalization and GELU activations. The state vector $\mathbf{s}_{t-1}$ is broadcast across all funds when computing per-fund outputs, enabling the same characteristic vector $\mathbf{x}_{i,t-1}$ to produce different portfolio weights and exposures under different market conditions without hand-crafted regime indicators.

ScoreNet and portfolio weight construction. The ScoreNet produces a raw score for each fund-factor pair:

$$\text{score}_{i,t-1}^{(k)} = h_\phi([\mathbf{x}_{i,t-1}; \mathbf{s}_{t-1}])_k, \quad k = 1, \dots, F, \quad (5)$$

where $[\cdot; \cdot]$ denotes concatenation. We convert scores to valid portfolio weights using two differentiable mappings, depending on the factor type.

Long-only factors ($k = 1, \dots, k_0$). Scores are passed through a cross-sectional softmax:

$$w_{i,t-1}^{(k)} = \frac{\exp(\text{score}_{i,t-1}^{(k)})}{\sum_{j=1}^N \exp(\text{score}_{j,t-1}^{(k)})}, \quad (6)$$

guaranteeing $w_{i,t-1}^{(k)} \geq 0$ and $\sum_i w_{i,t-1}^{(k)} = 1$ by construction. The resulting factor $f_t^{(k)}$ is the return of a fully-invested long-only portfolio.

Long-short market-neutral factors ($k = k_0 + 1, \dots, F$). We employ a softmax-minus-softmax construction:

$$w_{i,t-1}^{(k)} = \frac{\exp(\text{score}_{i,t-1}^{(k)})}{\sum_j \exp(\text{score}_{j,t-1}^{(k)})} - \frac{\exp(-\text{score}_{i,t-1}^{(k)})}{\sum_j \exp(-\text{score}_{j,t-1}^{(k)})}. \quad (7)$$

Since each softmax sums to one, market neutrality holds exactly: $\sum_{i=1}^N w_{i,t-1}^{(k)} = 1 - 1 = 0$. We further normalize by gross leverage, $\mathbf{w}_{t-1}^{(k)} \leftarrow \mathbf{w}_{t-1}^{(k)} / \sum_i |w_{i,t-1}^{(k)}|$, to stabilize factor scaling across training epochs. This construction is a differentiable analogue of “long the highest-score funds, short the lowest-score funds,” avoiding non-differentiable sorting operations while enforcing self-financing structure.

The final weight matrix concatenates long-only and long-short columns:

$$\mathbf{W}_{t-1} = [\mathbf{W}_{t-1}^{\text{LO}} \quad \mathbf{W}_{t-1}^{\text{LS}}] \in \mathbb{R}^{N \times F}. \quad (8)$$

In our experiments, we use $k_0 = 2$ long-only factors and $F - k_0 = 3$ long-short factors.

BetaNet. For each fund i , the BetaNet outputs conditional factor loadings:

$$\boldsymbol{\beta}_{i,t-1} = b_\theta([\mathbf{x}_{i,t-1}; \mathbf{s}_{t-1}]) \in \mathbb{R}^F, \quad (9)$$

stacked row-wise into the loading matrix $\mathbf{B}_{t-1} \in \mathbb{R}^{N \times F}$. These loadings are the model’s analogue of classical factor betas: $\beta_{i,t-1}^{(k)}$ measures fund i ’s exposure to factor k , conditional on current holdings and market state, enabling risk attribution and portfolio construction.

AlphaNet. The AlphaNet produces fund-specific intercepts:

$$\alpha_{i,t-1} = a_\eta([\mathbf{x}_{i,t-1}; \mathbf{s}_{t-1}]) \in \mathbb{R}, \quad (10)$$

capturing residual expected return not explained by factor exposures.

Reconstruction. Given weights $\mathbf{W}_{t-1}$ and realized returns $\mathbf{r}_t$, factor returns and reconstructed returns are computed as:

$$\mathbf{f}_t = \mathbf{W}_{t-1}^\top \mathbf{r}_t \in \mathbb{R}^F, \quad \hat{\mathbf{r}}_t = \boldsymbol{\alpha}_{t-1} + \mathbf{B}_{t-1} \mathbf{f}_t \in \mathbb{R}^N. \quad (11)$$

The bottleneck $\mathbf{f}_t$ has dimension $F \ll N$; training encourages portfolio weights that produce factor returns maximally informative for reconstructing the entire cross-section through the linear decoder $\mathbf{B}_{t-1}$.

3.4 Training Objective

Let $\Theta = (\psi, \phi, \theta, \eta)$ denote all learnable parameters. We minimize the composite objective:

$$\mathcal{L}(\Theta) = \mathbb{E}_t[\ell_{\text{rec}}(t) + \lambda_{\text{orth}} \ell_{\text{orth}}(t) + \lambda_\beta \ell_\beta(t) + \lambda_w \ell_w(t)], \quad (12)$$

where expectations are approximated by mini-batch averages over time indices.

Reconstruction loss. The primary objective is cross-sectional mean squared error:

$$\ell_{\text{rec}}(t) = \frac{1}{N} \sum_{i=1}^N (\hat{r}_{i,t} - r_{i,t})^2. \quad (13)$$

Orthogonality penalty. Without explicit encouragement, multiple factor portfolios may collapse onto the same dominant direction (e.g., a market-like portfolio). Let $\hat{\boldsymbol{\Sigma}}_f$ denote the sample covariance matrix of factor returns $\{\mathbf{f}_t\}$ within a mini-batch. We penalize off-diagonal mass:

$$\ell_{\text{orth}} = \left\| \hat{\boldsymbol{\Sigma}}_f - \text{diag}(\hat{\boldsymbol{\Sigma}}_f) \right\|_F^2, \quad (14)$$

encouraging the model to discover distinct sources of variation.

Regularization. We add ℓ_2 penalties on factor loadings and portfolio weights to control model capacity and prevent extreme concentrations:

$$\ell_\beta(t) = \|\mathbf{B}_{t-1}\|_F^2, \quad \ell_w(t) = \|\mathbf{W}_{t-1}\|_F^2. \quad (15)$$

These penalties shrink extreme exposures and overly concentrated portfolio weights, improving out-of-sample robustness.

Optimization. We train with Adam [21] using a learning rate of 10^{-2} and batch size 128 for 150 epochs. The best model state is selected by validation loss (computed on the last 10% of dates) with early stopping. Full hyperparameter details are in Appendix A.

3.5 Discussion: Why the Portfolio Bottleneck Matters

The portfolio bottleneck is what distinguishes the AFM from standard conditional autoencoders for asset pricing. In the model of Gu et al. [15], latent factors $\mathbf{f}_t$ are free variables produced by the encoder—they have no direct portfolio interpretation and cannot be traded without an additional projection step. By contrast, our constraint $\mathbf{f}_t = \mathbf{W}_{t-1}^\top \mathbf{r}_t$ provides three advantages.

Tradability. Each factor $f_t^{(k)} = \mathbf{w}_{t-1}^{(k)\top} \mathbf{r}_t$ is the realized return of a portfolio with known, pre-determined weights. A practitioner can implement these factors as live trading strategies—long-only or market-neutral—without any post-hoc approximation.

Interpretability. The portfolio weights $\mathbf{w}_{t-1}^{(k)}$ reveal which funds drive each factor: inspecting the largest weights identifies the economic themes each factor captures. The conditional loadings $\beta_{i,t-1}$ provide complementary fund-level exposure diagnostics for risk attribution.

Implicit regularization. Restricting factors to the span of realizable portfolio returns constrains the model’s hypothesis class relative to free latent codes. The model cannot exploit arbitrary rotations of the latent space to minimize reconstruction error; it must find directions in *portfolio-return space* that are genuinely informative for the cross-section. This connection echoes the classical theory of mimicking portfolios [16], where any priced factor admits a portfolio representation—the AFM learns such portfolios directly and nonlinearly. The portfolio bottleneck also relates to differentiable portfolio optimization [1, 8], but targets cross-sectional explanatory power rather than portfolio performance. The softmax weight construction can be viewed as a differentiable relaxation of parametric portfolio policies [4], extended from a single-portfolio objective to a multi-factor autoencoding framework.

4 Experiments

We evaluate the AFM on a broad universe of US mutual funds spanning 10 asset classes. Our experiments address four questions: (1) Does the portfolio bottleneck autoencoder outperform traditional factor models? (2) How does architecture complexity affect generalization? (3) Does multi-asset training improve performance via cross-asset transfer? (4) Where do the largest gains arise, and where do current limitations remain?

4.1 Experimental Setup

Data. We construct a dataset of approximately 1,000 US mutual funds drawn from 10 Morningstar categories: large cap, mid cap, small cap, emerging market (EM) equity, non-US equity, US REIT, US aggregate bonds, US high yield debt, EM debt, and municipal debt. Daily excess returns are computed over the risk-free rate. Fund-level characteristics—economic sector holdings (e.g., technology, healthcare, energy) and asset class allocations (e.g., equity, fixed income, cash)—are extracted from regulatory filings (SEC N-PORT) and cross-sectionally normalized per date. This characteristic set constitutes the alternative data that conditions both portfolio weights and factor loadings.

Evaluation metric. We use cross-sectional out-of-sample (OOS) R^2 as the primary metric. Returns are split temporally: the last 10% of dates form the held-out evaluation period, ensuring strict separation of training and test data. For each fund i , we compute the OOS R^2 of the model’s fitted returns $\hat{r}_{i,t}$ against realized returns $r_{i,t}$ over the evaluation period. We report the average and median OOS R^2 across funds, as well as per-category breakdowns.

Table 2: **Out-of-sample R^2 (%) across models and asset classes.** Each cell reports the average OOS R^2 for funds in that category. Higher is better. Bold indicates the best result per category. The AFM MLP All model achieves the highest average R^2 , with the largest gains on non-equity assets where Fama-French factors have near-zero explanatory power.

Category	AFM MLP All	AFM ResMLP All	AFM MLP Large Cap	AFM ResMLP Large Cap	FF 4-Factor
Large Cap	90	89	95	94	87
Mid Cap	89	86	88	86	85
Small Cap	86	83	84	82	88
EM Equity	80	83	48	48	43
Non-US Equity	85	77	69	67	62
US REIT	66	65	63	66	55
US High Yield	53	63	47	43	35
US Agg Bonds	84	80	11	17	11
EM Debt	44	44	14	13	11
Municipal Debt	27	30	5	7	4
Average	85.0	82.7	73.0	70.7	68.4
Median	88.1	85.4	86.0	81.5	75.4

Baseline. We compare against the Fama-French 4-factor model [5, 12], which includes the market (MKT-RF), size (SMB), value (HML), and momentum (UMD) factors. Factor loadings are estimated via OLS on the training period and applied to the held-out dates. This is the standard benchmark for cross-sectional return explanation and provides a strong baseline on equity categories.

Model configurations. We evaluate four AFM variants that vary along two dimensions—architecture and training universe:

- **AFM MLP All:** single-layer MLP backbone (hidden dim 128), trained on all 10 asset classes (34.8M parameters).
- **AFM ResMLP All:** residual MLP backbone (128→64), trained on all 10 asset classes (69.8M parameters).
- **AFM MLP Large Cap:** single-layer MLP, trained on large cap funds only (13.8M parameters).
- **AFM ResMLP Large Cap:** residual MLP, trained on large cap funds only (27.7M parameters).

All AFM models extract $F = 5$ factors (2 long-only, 3 long-short market-neutral), use a lookback window of $L = 252$ trading days, state dimension $S = 16$, GELU activations, layer normalization, dropout of 0.3–0.5, and are trained for 150 epochs with Adam (learning rate 0.01, batch size 128). The orthogonality penalty weight ranges from $\lambda_{\text{orth}} = 10^{-5}$ to 10^{-2} . Full hyperparameter details are provided in Appendix A.

Compute. All experiments are conducted on a single NVIDIA GPU. Each AFM training run completes within several hours. The Fama-French baseline requires only OLS regression and runs in seconds.

4.2 Main Results

Table 2 presents OOS R^2 across all five models and 10 asset classes. The best AFM configuration (MLP backbone, all-asset training) achieves an average OOS R^2 of 85.0%, compared to 68.4% for the Fama-French 4-factor model—a 16.6 percentage point improvement.

Equity categories. All models perform well on equity categories. Large cap, mid cap, and small cap R^2 exceeds 82% for every model, including Fama-French. This is expected: the Fama-French factors are designed for US equities and capture the dominant sources of variation in these categories. Notably, the Fama-French model achieves 88% on small cap, slightly outperforming the AFM MLP All (86%), which may reflect the tight alignment between the SMB factor and small-cap return variation.

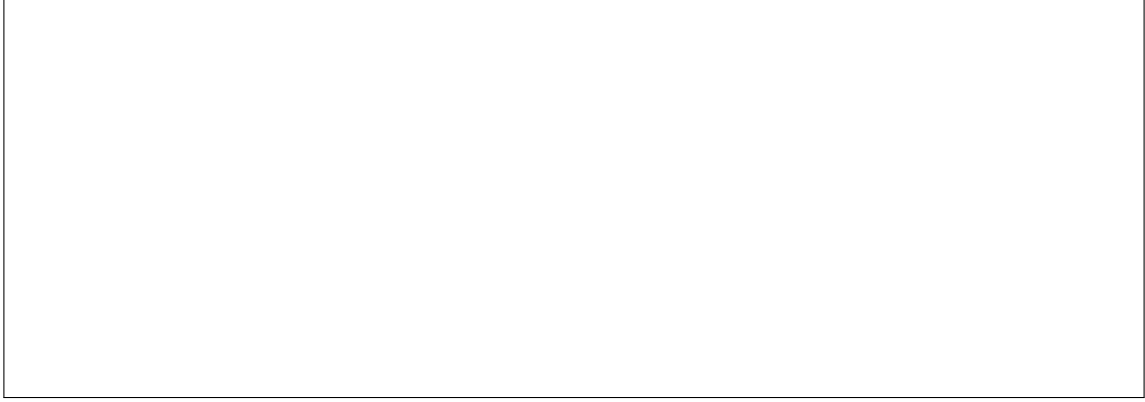


Figure 2: **OOS R^2 by asset class and model.** Grouped bar chart comparing all five models across 10 mutual fund categories. The AFM models trained on all asset classes (blue and orange bars) dominate on non-equity categories (right side), while all models perform comparably on equity categories (left side). Fama-French (gray) achieves near-zero R^2 on bonds and EM debt, where the AFM captures interest rate and credit factors through its portfolio bottleneck.

Table 3: **Ablation: architecture and training universe.** Each cell reports average OOS R^2 (%) and parameter count. The single-layer MLP consistently outperforms the deeper residual MLP, and multi-asset training (“All”) consistently outperforms large-cap-only training. Higher is better.

	All Assets	Large Cap Only
MLP (single layer)	85.0% (34.8M)	73.0% (13.8M)
Residual MLP (2 layers)	82.7% (69.8M)	70.7% (27.7M)

Non-equity categories. The AFM’s advantages emerge most clearly on asset classes where traditional equity factors have limited explanatory power. On US aggregate bonds, the AFM MLP All achieves 84% R^2 compared to just 11% for Fama-French—a 73 percentage point gap. Similarly, on EM debt (44% vs. 11%), US high yield (53% vs. 35%), and US REITs (66% vs. 55%), the AFM substantially outperforms. These categories are driven by interest rate, credit, and currency risk factors that equity-based models cannot capture. The portfolio bottleneck learns cross-asset factor portfolios that span these risk dimensions.

Figure 2 provides a visual comparison across all categories and models, highlighting the systematic advantage of multi-asset AFM training on non-equity categories.

4.3 Ablation Studies

We conduct two ablations to isolate the contributions of architecture choice and training universe composition. Table 3 summarizes the results in a 2×2 grid.

Architecture complexity. The single-layer MLP (34.8M parameters) outperforms the deeper residual MLP (69.8M parameters) by 2.3 percentage points in the all-asset setting (85.0% vs. 82.7%) and by 2.3 percentage points in the large-cap setting (73.0% vs. 70.7%). This result is consistent across both training universes, suggesting that the additional depth and skip connections of the residual architecture add capacity without useful expressiveness for this task. We hypothesize that mutual fund returns exhibit smoother cross-sectional structure than individual stock returns—funds are diversified portfolios, so their return cross-section is lower-dimensional—and a single hidden layer suffices to capture the relevant nonlinearities in the characteristics-to-loadings mapping.

Training universe. Multi-asset training yields a 12.0 percentage point improvement over large-cap-only training for the MLP backbone (85.0% vs. 73.0%) and a 12.0 percentage point improvement for the residual

MLP (82.7% vs. 70.7%). The gains are concentrated in non-equity categories: US aggregate bonds improve from 11% to 84%, EM debt from 14% to 44%, and non-US equity from 69% to 85%. Even equity categories benefit modestly—EM equity improves from 48% to 80%. This demonstrates cross-asset transfer learning: by training jointly across equities, bonds, REITs, and emerging markets, the model discovers shared factor structure (e.g., common interest rate and credit risk drivers) that single-asset training cannot identify. The portfolio bottleneck is essential to this transfer—it forces the model to construct factor portfolios whose returns are informative for *all* asset classes simultaneously.

4.4 Cross-Asset Analysis

To better understand where the AFM’s gains originate, we group the 10 categories by asset type. *Equity categories* (large cap, mid cap, small cap, EM equity, non-US equity) achieve above 80% R^2 for all AFM all-asset models, with the AFM and Fama-French performing comparably on domestic equities. The differentiation between models comes primarily from three groups:

- **International and EM equity:** The AFM captures diversification and currency factors that US-centric models miss. EM equity improves from 43% (FF) to 80–83% (AFM All).
- **Fixed income:** US aggregate bonds (84% vs. 11%) and US high yield (53–63% vs. 35%) show the largest absolute gains. The portfolio bottleneck learns rate-sensitive and credit-sensitive factor portfolios that have no analogue in the Fama-French framework.
- **Alternative and EM debt:** US REITs (66% vs. 55%) and EM debt (44% vs. 11%) benefit from cross-asset training but remain below equity-level R^2 , reflecting the distinct risk drivers and thinner cross-sections in these categories.

Municipal debt remains the most challenging category, with R^2 peaking at 30%. We discuss this limitation in Section 4.6.

4.5 Training Dynamics

Figure 3 shows the training and validation loss trajectories for all four AFM variants. All models converge rapidly within 10–15 epochs, with continued modest improvement thereafter. The best validation loss is reached between epochs 84 and 114, depending on the configuration. Importantly, no overfitting is observed for any model: validation loss stabilizes and does not deteriorate even after 150 epochs of training. This suggests that the portfolio bottleneck acts as an effective implicit regularizer—by constraining the latent factors to be returns of explicit portfolios rather than free latent codes, the model’s representational capacity is disciplined in a way that prevents memorization of training-period return patterns.

4.6 Limitations

Several limitations of the current evaluation deserve emphasis. First, **municipal debt** R^2 peaks at 30%, likely because our fund-level characteristics (sector holdings and asset class allocations) do not capture the credit quality, duration, and tax-status features that drive municipal bond returns. Richer characteristic inputs—including credit ratings, effective duration, and state-level tax information—would likely improve performance on this category. Second, our evaluation is based on **explanatory** R^2 only; we do not assess portfolio-level performance metrics such as Sharpe ratios, turnover, or transaction costs, which are critical for practical deployment. Third, we report results on a **single temporal split** (last 10% of dates) rather than rolling or cross-validated evaluation, due to the computational cost of training. While the temporal split is standard practice in asset pricing [14, 15], future work should examine robustness across different evaluation windows.

5 Conclusion

We proposed the Autoencoding Factor Model (AFM), a neural autoencoder with a portfolio bottleneck that constrains each latent factor to be the return of an explicit, tradable portfolio. Portfolio weights and conditional

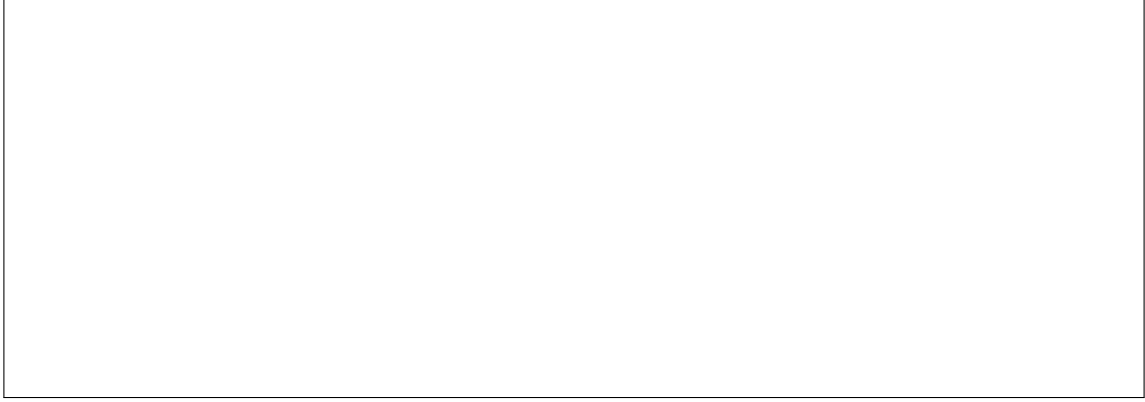


Figure 3: **Training loss trajectories for all AFM variants.** Epoch (x-axis) vs. reconstruction loss (y-axis) for training (solid) and validation (dashed) sets. All four configurations—MLP All (blue), ResMLP All (orange), MLP Large Cap (green), ResMLP Large Cap (red)—converge within 10–15 epochs and show no signs of overfitting through 150 epochs. Best validation loss is reached between epochs 84–114.

factor loadings are produced by neural networks conditioned on mutual fund characteristics—economic sector holdings and asset class allocations derived from regulatory filings—and a learned market-state embedding from past returns. The entire system is trained end-to-end via reconstruction loss, jointly learning portfolio construction, factor loadings, and asset intercepts.

Trained on approximately 1,000 US mutual funds spanning 10 asset classes, the AFM with 5 factors achieves 85.0% average out-of-sample R^2 , a 16.6 percentage point improvement over the Fama–French 4-factor model. The largest gains arise on non-equity assets—US aggregate bonds (84% vs. 11%), emerging market debt (44% vs. 11%), and US high yield (53% vs. 35%)—where traditional equity-based factors have near-zero explanatory power. Ablation studies demonstrate that simpler architectures (single-layer MLP) outperform deeper alternatives (residual MLP) and that multi-asset training is critical for cross-asset generalization, enabling transfer learning of shared factor structure across equities, bonds, and REITs.

Limitations. Several limitations warrant discussion. First, our evaluation is limited to explanatory R^2 ; we do not assess portfolio-level performance metrics such as Sharpe ratios, turnover, or transaction costs, which are essential for practical deployment. Second, municipal debt ($R^2 \approx 30\%$) and emerging market debt ($R^2 \approx 44\%$) remain challenging, suggesting that the current fund-level characteristics are insufficient for these asset classes—richer inputs such as credit quality, duration, and tax status may be needed. Third, results are based on a single train/validation split; cross-validation would strengthen robustness claims but is computationally prohibitive at present. Finally, the model lacks formal theoretical guarantees on factor recovery or out-of-sample generalization.

Future work. Several directions are promising. Incorporating richer characteristics—credit ratings, duration profiles, macroeconomic indicators—may improve performance on fixed-income assets. Extending the framework to individual securities (stocks and bonds) rather than mutual funds would test scalability to larger cross-sections. A variational autoencoder (VAE) extension [9] could provide uncertainty quantification over factor returns and loadings. Portfolio-level evaluation, including live trading simulation with realistic transaction costs, would bridge the gap between explanatory power and economic value. Finally, theoretical analysis of the portfolio bottleneck as an implicit regularizer—characterizing when and why it improves generalization relative to free latent codes—remains an open and important question.

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A Implementation Details

Hyperparameters. Table 4 provides the full hyperparameter configuration used across all AFM experiments.

Table 4: **Hyperparameter settings for all AFM experiments.** Unless otherwise noted, all values are shared across the four AFM variants.

Hyperparameter	Value
Number of factors (F)	5 (2 long-only, 3 long-short)
Lookback window (L)	252 trading days
State dimension (S)	16
Hidden dimension (MLP)	128
Hidden dimensions (Residual MLP)	128 $\rightarrow$ 64
Activation function	GELU
Normalization	Layer normalization
Dropout rate	0.3–0.5
Optimizer	Adam
Learning rate	0.01
Batch size	128 (time indices)
Training epochs	150 (with early stopping)
Orthogonality penalty (λ_{orth})	10^{-5} to 10^{-2}
Loading regularization (λ_{β})	L2 penalty
Weight regularization (λ_w)	L2 penalty
Portfolio weight construction	
Long-only factors	Softmax over scores
Long-short factors	Softmax-minus-softmax with gross leverage normalization
Evaluation	
Train/test split	Last 10% of dates held out
Metric	Per-fund OOS R^2 , averaged across funds

Parameter counts. Table 5 reports the parameter counts for each model configuration. The parameter count varies primarily with the backbone architecture and the number of assets in the training universe (which determines the dimensionality of the ReturnEncoder input).

Table 5: **Parameter counts by model configuration.**

Model	Backbone	Universe	Parameters
AFM MLP All	MLP (128)	All 10 classes	34.8M
AFM ResMLP All	ResMLP (128→64)	All 10 classes	69.8M
AFM MLP Large Cap	MLP (128)	Large cap only	13.8M
AFM ResMLP Large Cap	ResMLP (128→64)	Large cap only	27.7M
Fama-French 4-Factor	N/A (OLS)	Large cap	0

Software and hardware. All AFM models are implemented in PyTorch. Training is performed on a single NVIDIA GPU. Each full training run (150 epochs) completes within several hours. The Fama-French baseline is implemented via ordinary least squares regression using standard linear algebra routines.

Characteristic preprocessing. Fund-level characteristics are extracted from SEC N-PORT filings and include two groups: (i) economic sector holdings (percentage of fund AUM allocated to each GICS sector) and (ii) asset class allocations (percentage in equities, fixed income, cash, alternatives, etc.). All characteristics are cross-sectionally normalized per date to have zero mean and unit variance across the fund universe.

B Additional Per-Category Results

Median OOS R^2 . Table 6 reports the median (rather than mean) OOS R^2 across funds. The median is more robust to outlier funds and provides a complementary view of typical model performance. The ranking of models is preserved: AFM MLP All achieves the highest median R^2 at 88.1%, compared to 75.4% for Fama-French.

Table 6: **Median OOS R^2 (%) across models.** The median is computed across all funds in each model’s evaluation universe. Higher is better.

Model	Median OOS R^2
AFM MLP All	88.1%
AFM ResMLP All	85.4%
AFM MLP Large Cap	86.0%
AFM ResMLP Large Cap	81.5%
Fama-French 4-Factor	75.4%

Best epoch by configuration. Table 7 reports the epoch at which each AFM variant achieves its best validation loss. All models converge well before the 150-epoch training budget, and the best epochs cluster in the 84–114 range.

Table 7: **Best validation epoch by model configuration.**

Model	Best Epoch
AFM MLP All	84
AFM ResMLP All	100
AFM MLP Large Cap	112
AFM ResMLP Large Cap	114